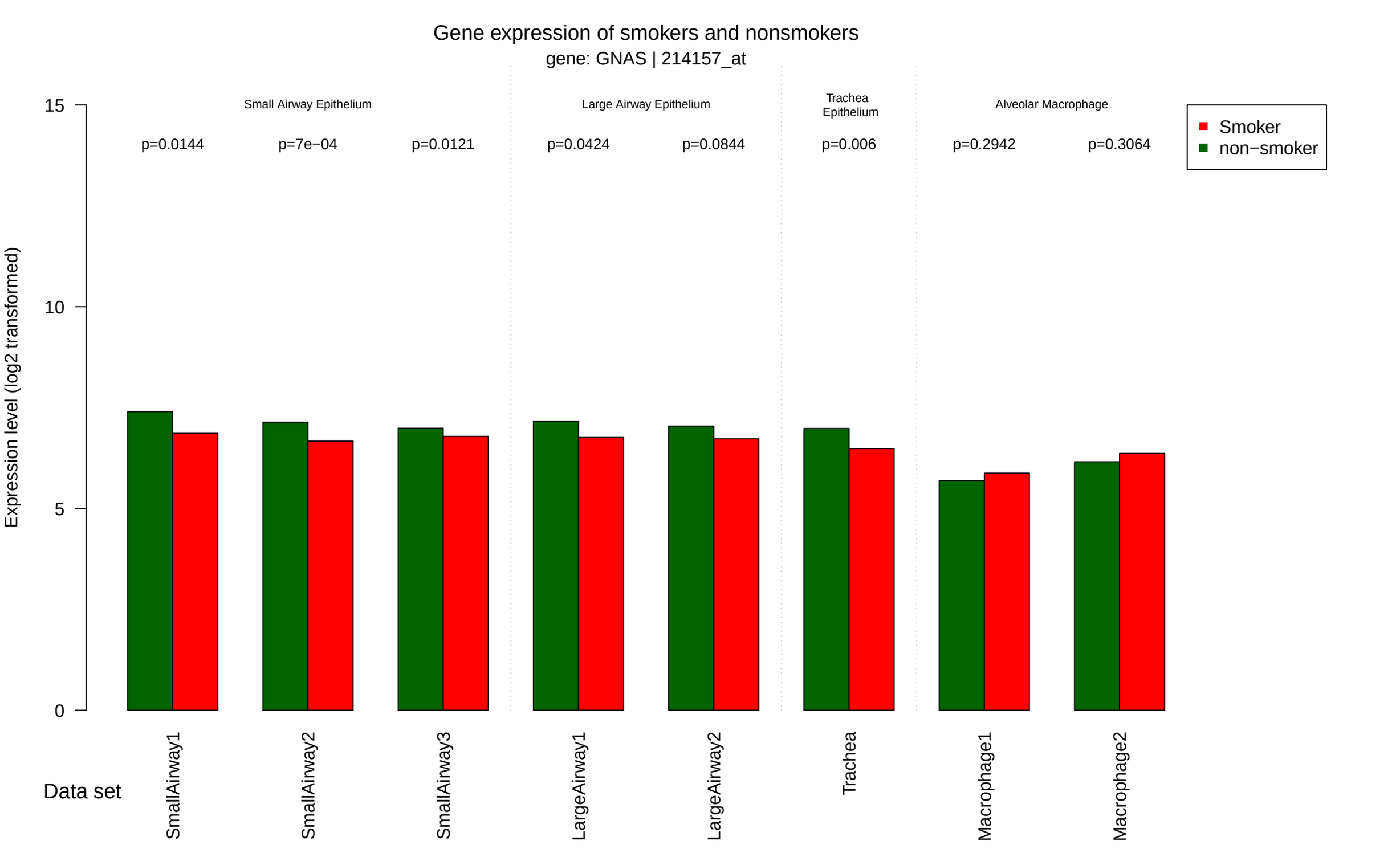
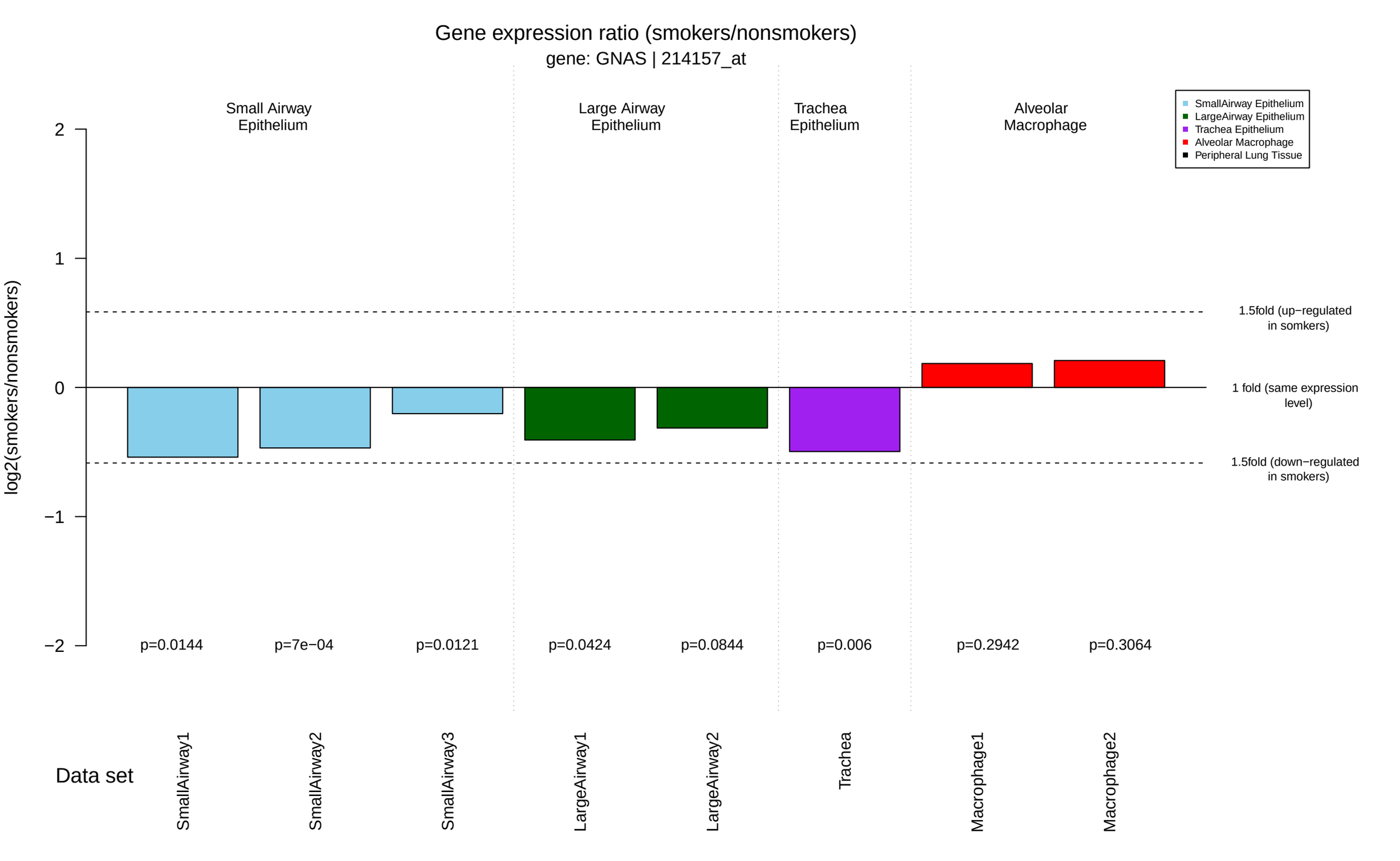
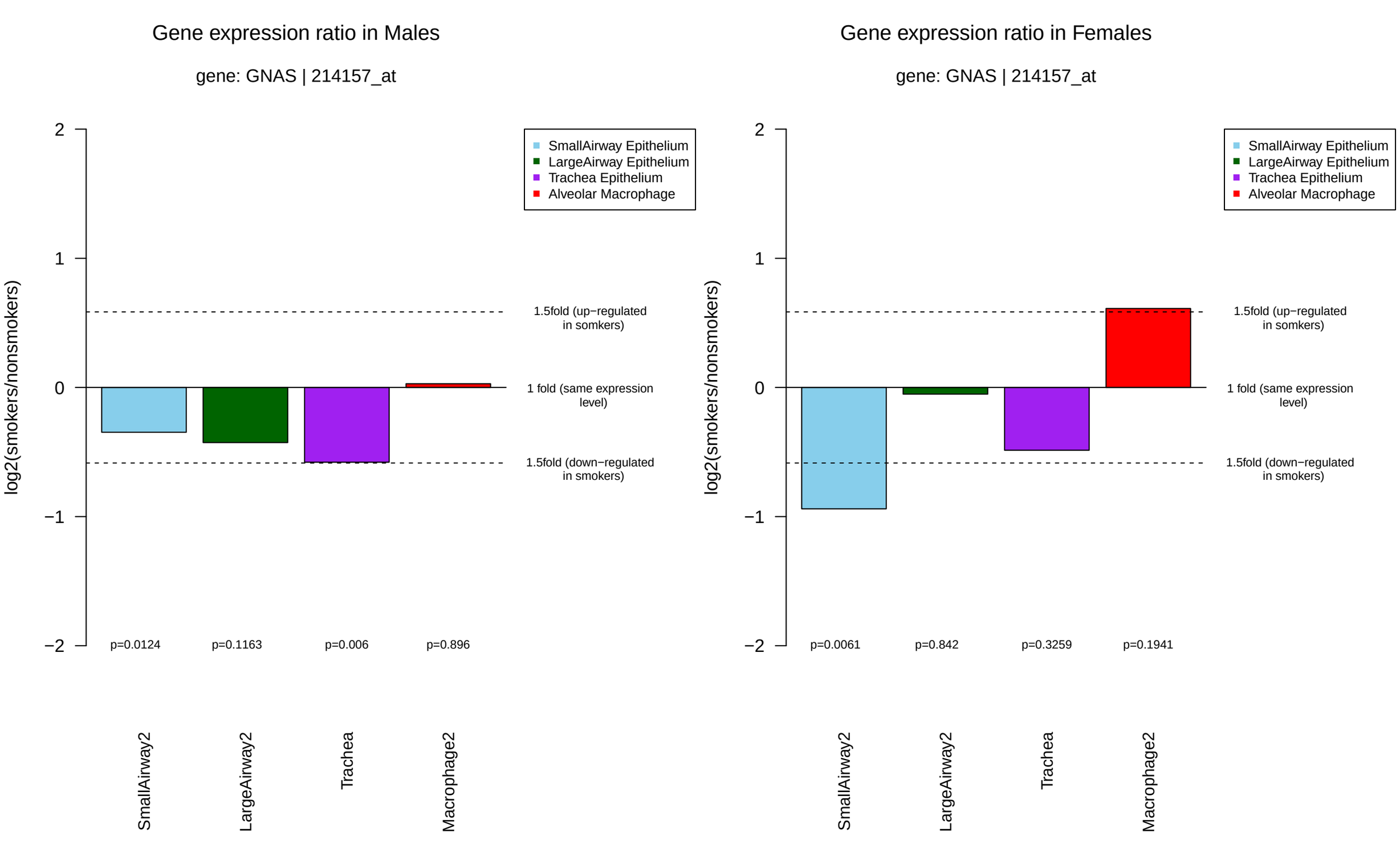
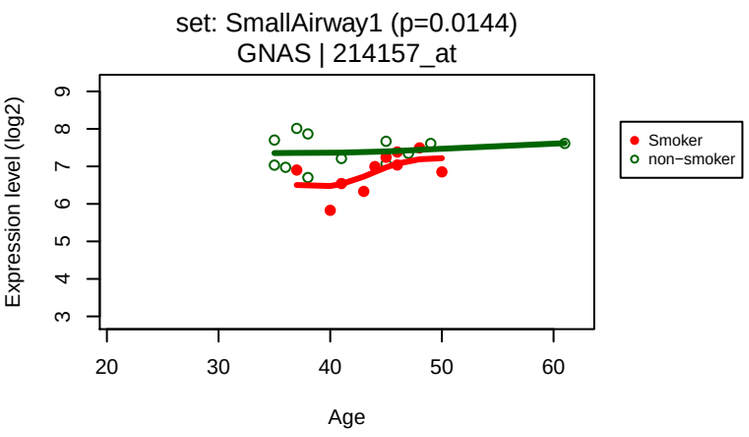
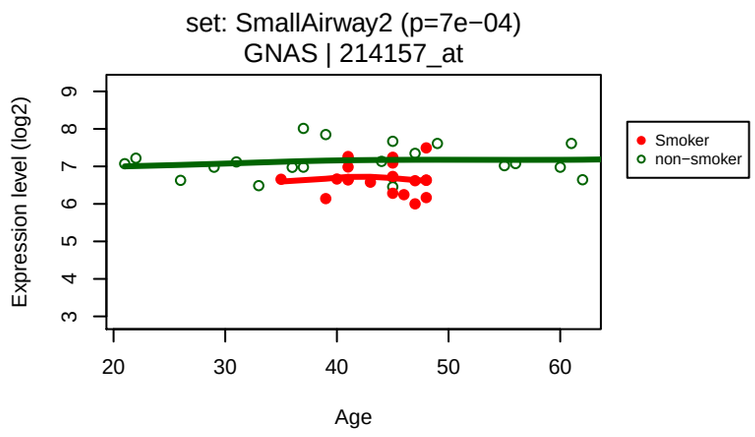
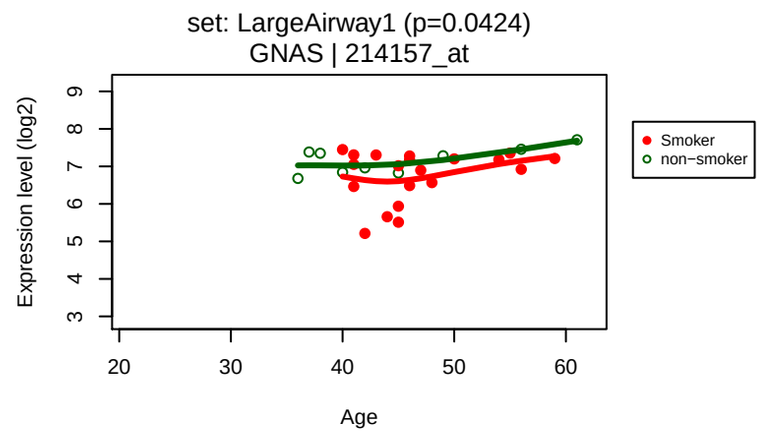

Gene expression of smokers and nonsmokers
gene: GNAS | 214157_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0144
p=7e−04
p=0.0121
p=0.0424
p=0.0844
p=0.006
p=0.2942
p=0.3064
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

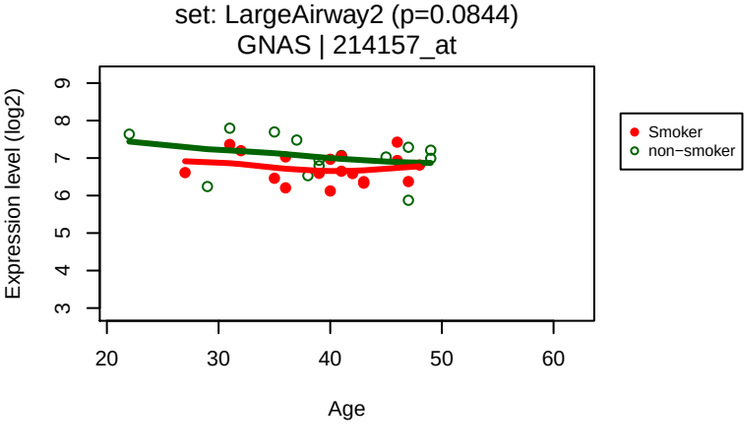

Gene expression ratio (smokers/nonsmokers)
gene: GNAS | 214157_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0144
p=7e−04
p=0.0121
p=0.0424
p=0.0844
p=0.006
p=0.2942
p=0.3064
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

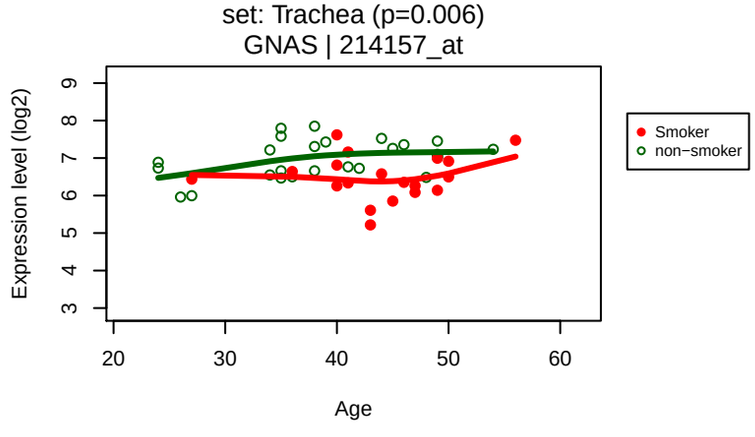

Gene expression ratio in Males
gene: GNAS | 214157_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0124
p=0.1163
p=0.006
p=0.896
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNAS | 214157_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0061
p=0.842
p=0.3259
p=0.1941
SmallAirway2
LargeAirway2
Trachea
Macrophage2

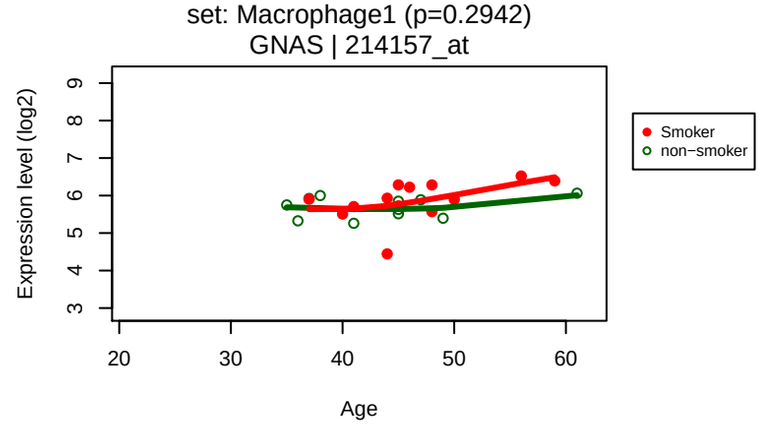

set: SmallAirway1 (p=0.0144)
GNAS | 214157_at
Expression level (log2)
Age
Smoker
non−smoker

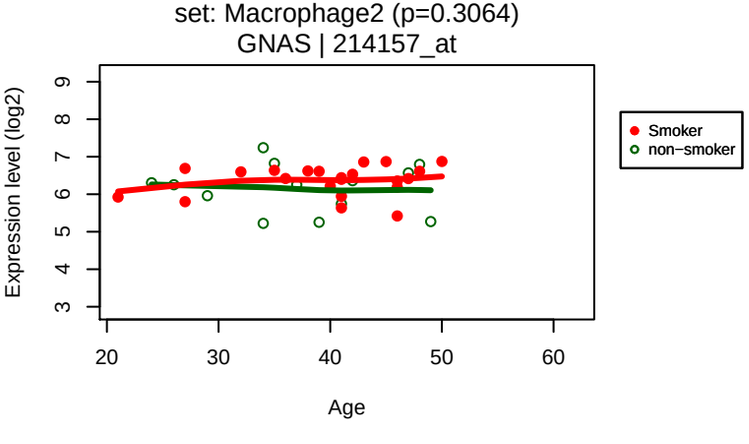

set: SmallAirway2 (p=7e−04)
GNAS | 214157_at
Expression level (log2)
Age
Smoker
non−smoker

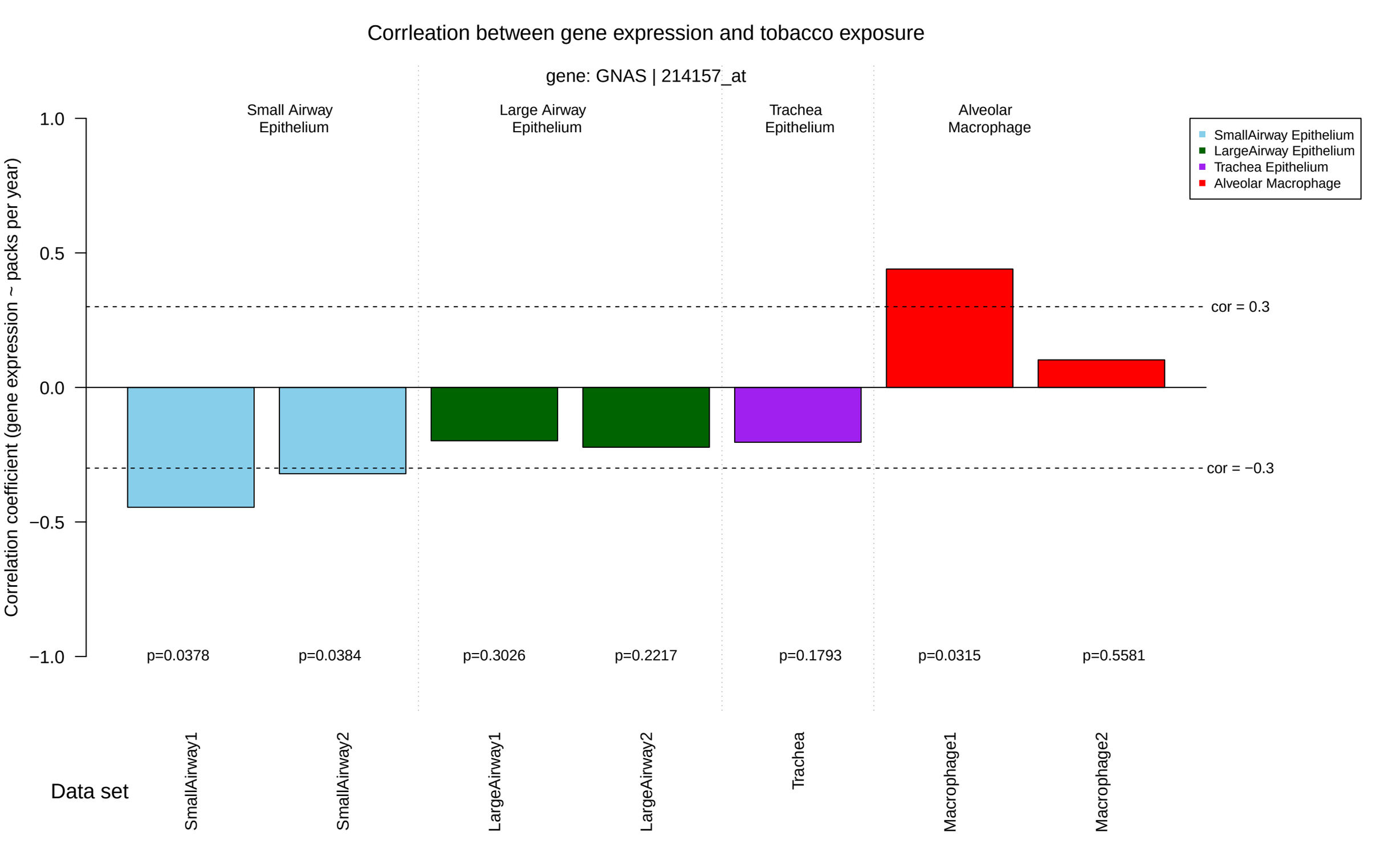

set: LargeAirway1 (p=0.0424)
GNAS | 214157_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0844)
GNAS | 214157_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.006)
GNAS | 214157_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2942)
GNAS | 214157_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3064)
GNAS | 214157_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNAS | 214157_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0378
p=0.0384
p=0.3026
p=0.2217
p=0.1793
p=0.0315
p=0.5581
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2